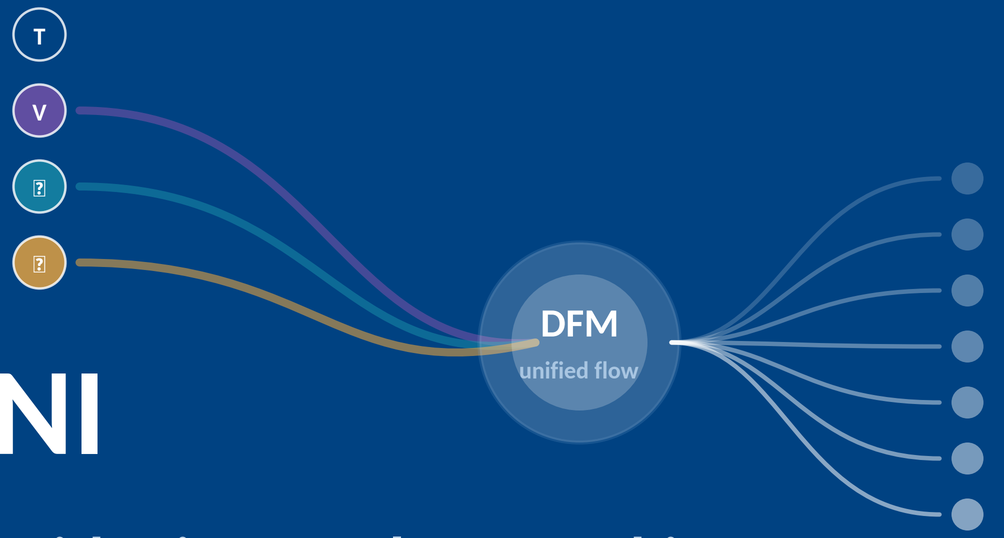


NExT-OMNI

Any-to-Any Omnimodal Foundation Models with Discrete Flow Matching

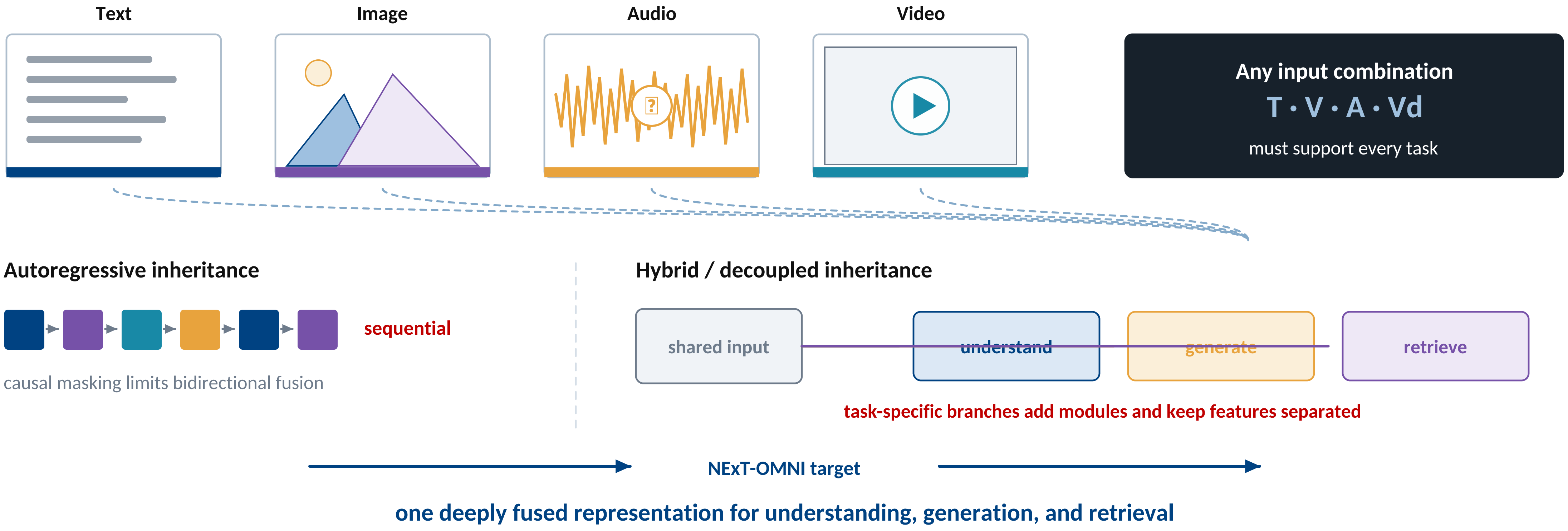
any input → any output

Run Luo · Xiaobo Xia · Lu Wang · Longze Chen · Renke Shan · Jing Luo · Min Yang · Tat-Seng Chua

SIAT, Chinese Academy of Sciences · UCAS · NExT++ Research Center · NUS

arXiv:2510.13721v2 [cs.CL]

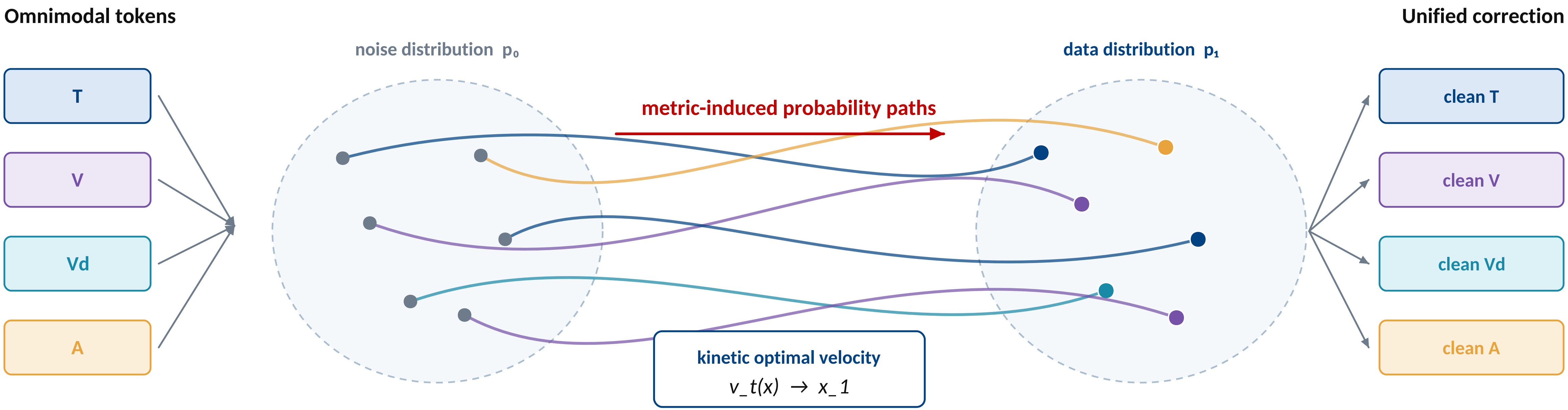
The conflict — one model, three jobs, four modalities



Autoregression is sequential; hybrid systems unify tasks by splitting the model.

Introduction and Figure 1 of the paper.

The alternative — discrete flow learns to correct in parallel



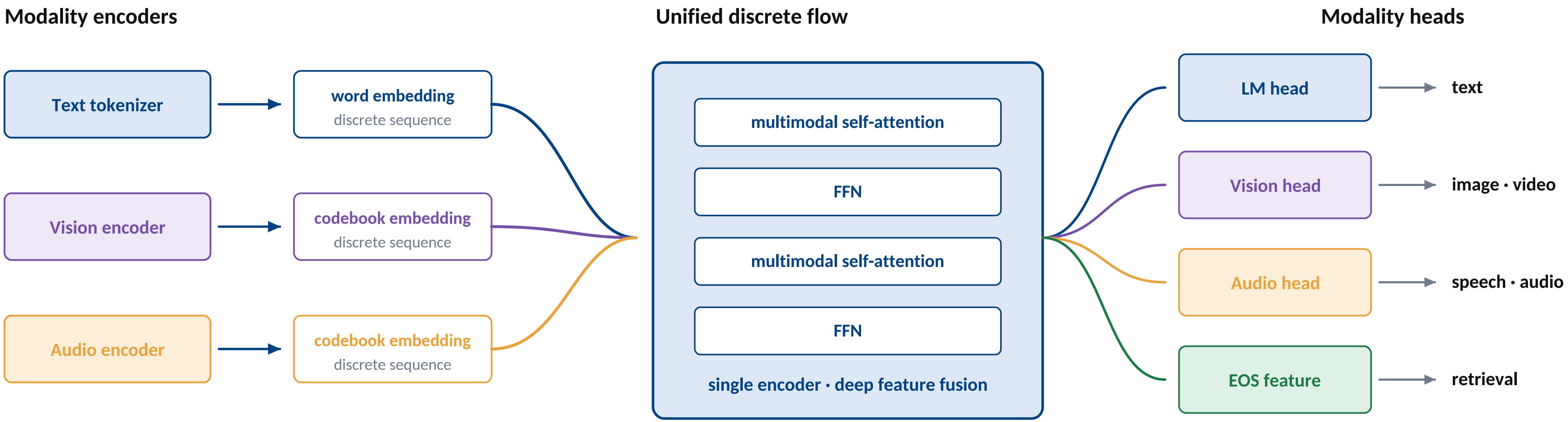
$$L\_CE = E_{t,x\_1,x\_t}\,[\,-\,\Sigma\_i\ \log\ p_{1|t}(x^\wedge i\_1 \mid x\_t)\,]$$

every position is predicted from the entire noisy multimodal sequence

Corrupt globally, predict globally, refine iteratively — with bidirectional context.

Sections 2.3–2.4 and Appendix A of the paper.

NExT-OMNI — a single deeply fused representation



Representation warmup $L^{M}_{total} = L^{M}_{rec} + L^{M}_{sem}$

reconstruction preserves detail · semantics align modalities

Joint flow training $L = \lambda_1 L_{CE} + \lambda_2 L^{V}_{rec} + \lambda_3 L^{A}_{rec}$

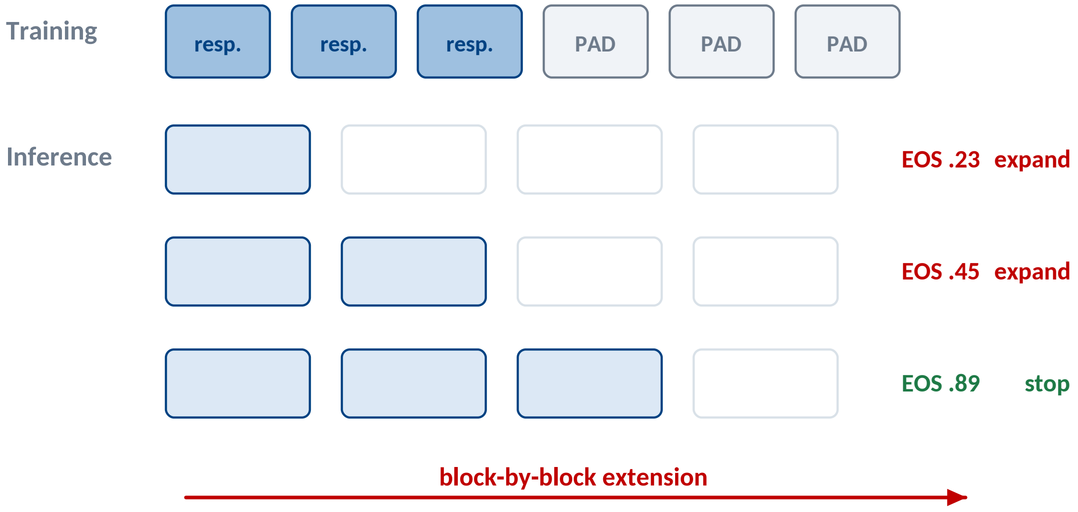
reconstruction feedback prevents loss of fine-grained information

One backbone serves understanding, generation, and retrieval — no task router required.

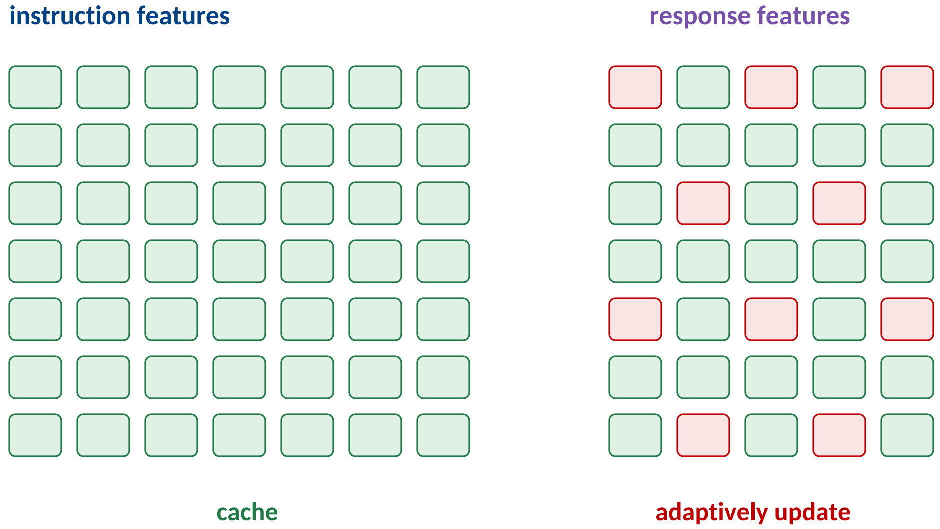
Sections 2.1–2.3 and Figure 2 of the paper.

Training and inference — make iterative decoding practical

Dynamic-length generation



Vanilla adaptive cache



1.4× higher training efficiency

1.2× faster response than AR

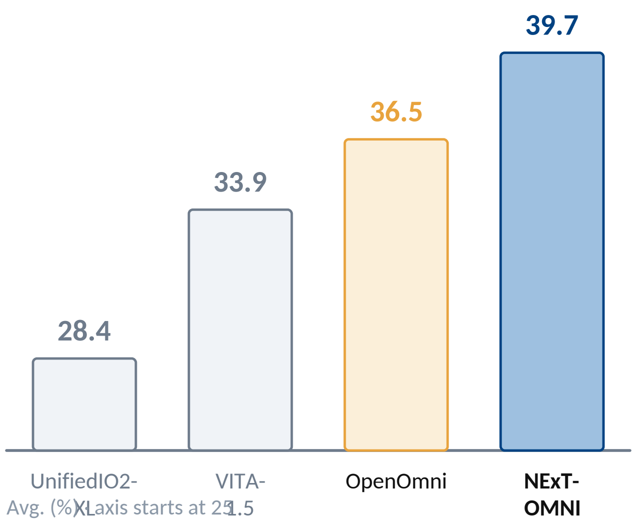
Interleave modalities, grow responses by blocks, and cache features that barely change.

Section 2.4 and Figure 3 of the paper.

Evidence — the unified flow transfers across task families

Omnimodal understanding

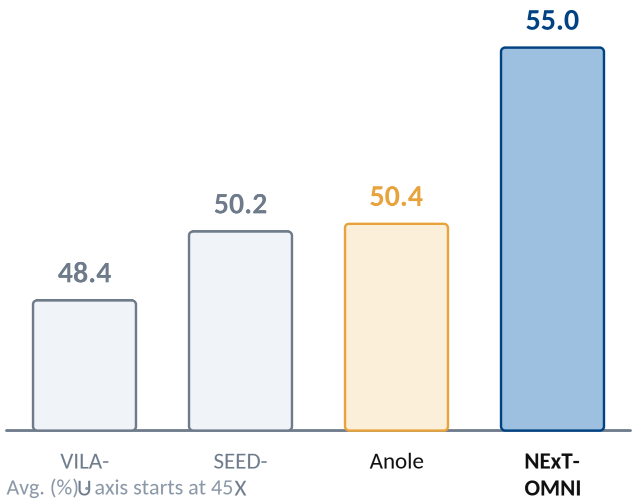
Table 1



NExT-OMNI +3.2

Multi-turn vision

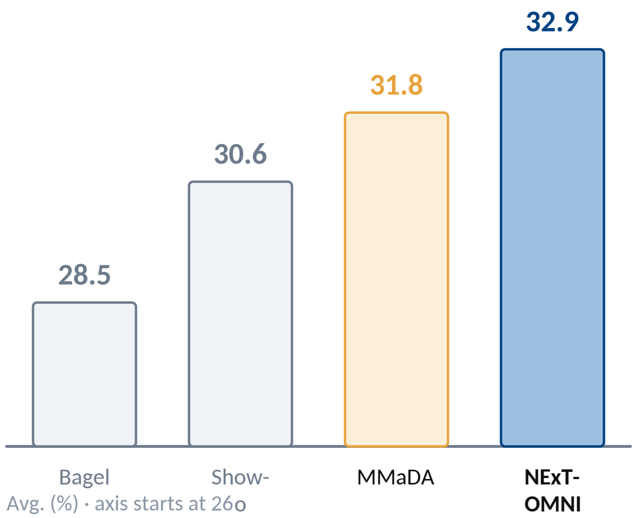
Table 2



NExT-OMNI +4.6

Cross-modal retrieval

Table 4



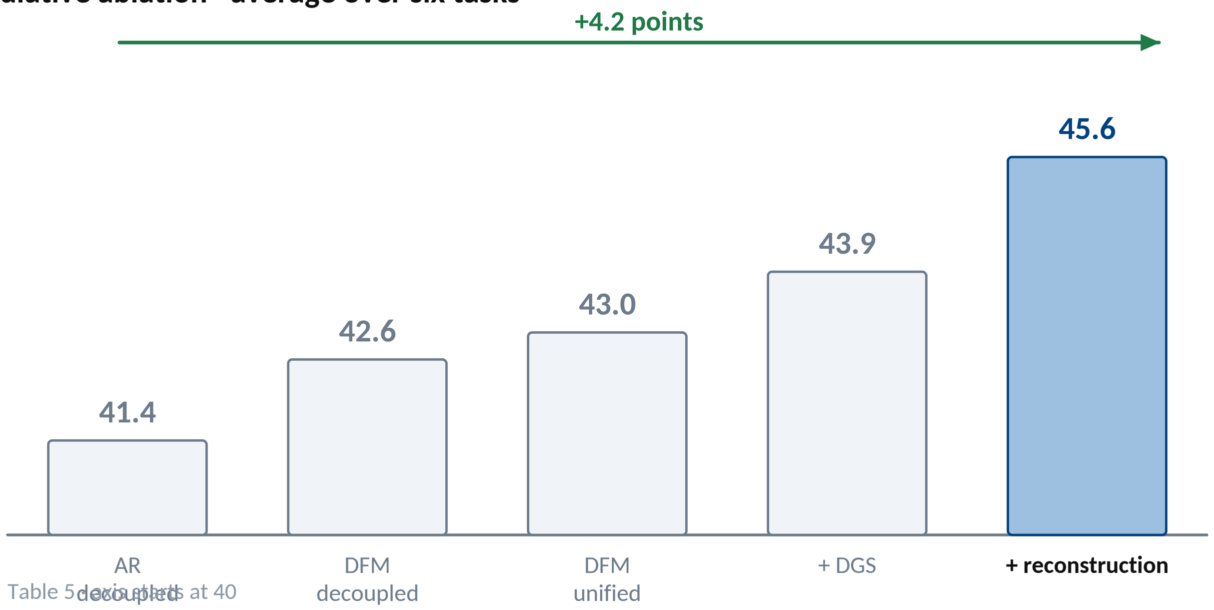
NExT-OMNI +1.1

NExT-OMNI leads on understanding, multi-turn vision, and cross-modal retrieval.

Main-paper Tables 1, 2, and 4.

What the ablation says — unification needs the full recipe

Cumulative ablation · average over six tasks



What each component resolves

1. **DFM**
   replace causal decoding
2. **Unified rep.**
   support retrieval
3. **Dynamic length**
   recover understanding
4. **Reconstruction**
   retain fine detail

NEXT **action trajectories · physical-AI video**

BOUNDARY benchmark breadth ≠ deployed safety

DFM opens the door; dynamic length and reconstruction make the unified model work.

Run Luo, Xiaobo Xia, Lu Wang, Longze Chen, Renke Shan, Jing Luo, Min Yang, and Tat-Seng Chua. NExT-OMNI: Towards Any-to-Any Omnimodal Foundation Models with Discrete Flow Matching. arXiv:2510.13721v2, 2025.